

Day 94: Basic Probability Distributions

1. Day number

Day 94

2. Topic name

Probability Distributions and the Normal Distribution

A probability distribution describes **how possible values are spread out and how likely different values are**.

3. Connection

Yesterday, you learned the probability of individual events, such as:

$$P(\text{Heads}) = 0.5$$

Today, instead of asking about only one event, you will look at **many possible values together**.

A simple connection is:

Probability
↓
How likely is one event?

Probability distribution
↓
How is probability spread across all possible values?

4. Important topics

Today you will learn:

- **Random variable** idea
- **Probability distribution**
- **Discrete** values
- **Continuous** values
- **Normal distribution**
- **Bell curve**

The main goal is conceptual understanding rather than formulas.

5. Foundational notes

Suppose you roll a fair six-sided die.

The possible results are:

1, 2, 3, 4, 5, 6

Each value has some probability.

For a fair die:

1 → 1/6
2 → 1/6
3 → 1/6
4 → 1/6
5 → 1/6
6 → 1/6

Instead of studying these probabilities separately, we can describe all of them together as a **probability distribution**.

Think of a distribution as answering:

| Which values are possible, and how common or likely is each part of the range?

6. Random variable idea

A **random variable** is simply a numerical value whose result depends on some uncertain process.

For example, suppose:

X = result of rolling a die

Then X could become:

1, 2, 3, 4, 5, or 6

Another example:

X = number of customers entering a shop in one hour

Possible values might be:

0, 1, 2, 3, 4, ...

You do not need advanced notation yet.

Just remember:

Random process



Produces a numerical result



That result can be treated as a random variable

7. What is a distribution?

Imagine recording the scores of many students:

45, 52, 55, 58, 60, 60, 61, 62, 65, 69, 75

Some areas contain **many values**, while other areas contain only a few.

A distribution shows this overall pattern.

For example:

Score area	Amount of data
40–49	few
50–59	some
60–69	many
70–79	few

You can think of a distribution as the **shape formed by the data**.

It can tell us whether values:

- gather around the center,
- spread widely,
- appear evenly,
- or concentrate more in certain areas.

8. Discrete vs continuous idea

Discrete values

Discrete values usually come from **counting** and have separate possible values.

For example:

Number of customers:

0, 1, 2, 3, 4, ...

You cannot normally have:

2.7 customers

Another example is a die:

1, 2, 3, 4, 5, 6

There are clear, separate possibilities.

Continuous values

Continuous values usually come from **measurement**.

For example, someone's height could be:

170 cm
170.2 cm
170.25 cm
170.253 cm
...

There can be many possible values between two measurements.

A useful shortcut is:

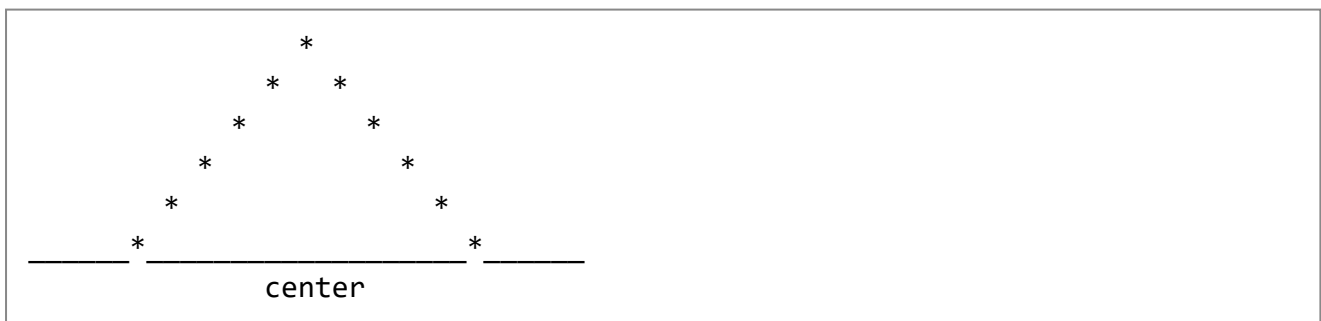
Discrete → usually counting
Continuous → usually measuring

This is a beginner-friendly rule rather than a complete mathematical definition.

9. The normal distribution

One particularly important distribution is the **normal distribution**.

Its shape looks approximately like a bell:



Because of this appearance, it is commonly called a **bell curve**.

10. Bell curve intuition

In a normal distribution, values tend to appear **more often near the center**.

Values farther from the center become progressively less common.

Imagine exam scores centered around 70.

You might see many scores such as:

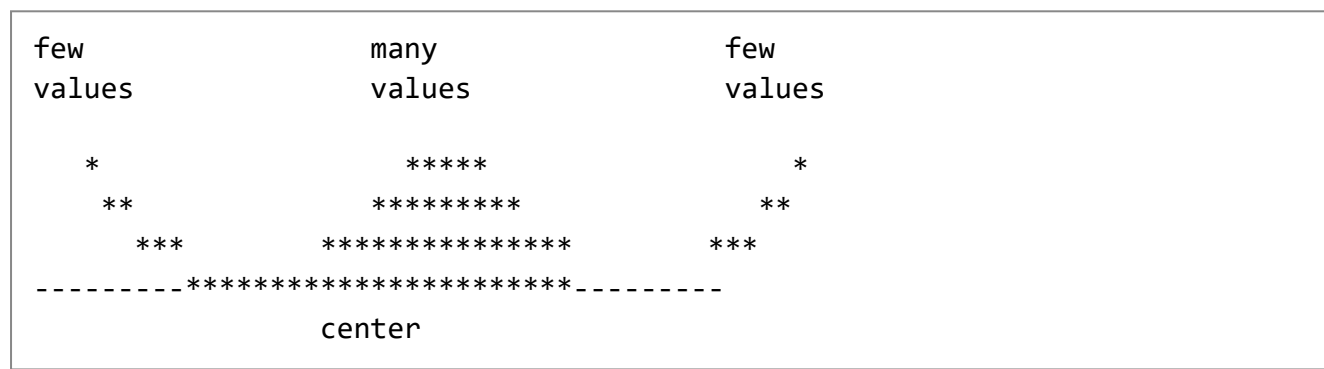
67, 68, 69, 70, 70, 71, 72, 73

You might see fewer scores such as:

50
90

And extremely low or extremely high scores might be even less common.

Conceptually:



The tallest part represents the region where values are most concentrated.

11. Easy example

Suppose we record these measurements:

48, 49, 50, 50, 50, 51, 52

Notice what happens.

Near the center:

49, 50, 50, 50, 51

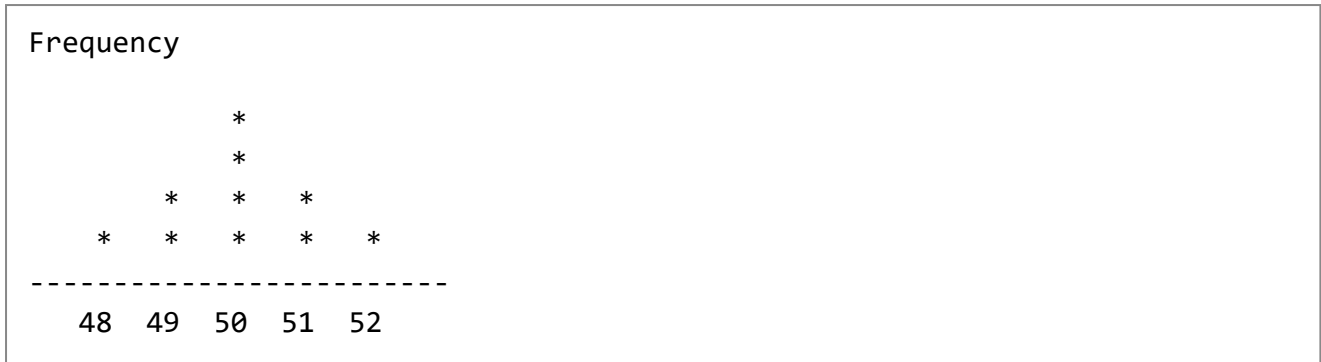
there are many observations.

Farther away:

48, 52

there are fewer.

Conceptually, we might draw:



The values are concentrated around 50.

This is the basic intuition behind the center of a bell-shaped distribution.

It does **not** mean that every real dataset will form a perfect normal distribution.

12. Problem statement

Consider these values:

18, 19, 20, 20, 20, 21, 22

Without performing advanced calculations, answer conceptually:

1. Where is the approximate center?
2. Which values occur near the center?
3. Which values are farther from the center?
4. Where would the tallest part of a distribution appear?
5. Where would the smaller parts appear?
6. Does this tiny dataset look roughly centered or heavily shifted toward one side?

The goal is to **interpret the distribution**, not calculate an advanced probability.

13. Concepts used

This problem uses:

- numerical data
- random variable idea
- possible values
- frequency
- distribution
- center
- spread
- discrete vs continuous data
- normal distribution
- bell curve

It also connects with Day 92:

Mean
↓
helps describe center

Standard deviation
↓
helps describe spread

Those ideas become useful when describing distributions.

14. Thought process

When examining a simple distribution, think in this order:

What values do I have?
↓
Where are most values located?
↓
What appears to be the center?
↓
How far do values extend from the center?
↓
Are values common near the center?
↓
Are extreme values less common?
↓
What overall shape do I see?

You do not need to immediately ask:

| Is this mathematically a perfect normal distribution?

At this stage, focus on recognizing the general pattern.

15. Beginner-friendly visual explanation

Imagine the following values:

1, 2, 3, 3, 3, 4, 5

We could stack them according to how often they appear:

•
•
• • •

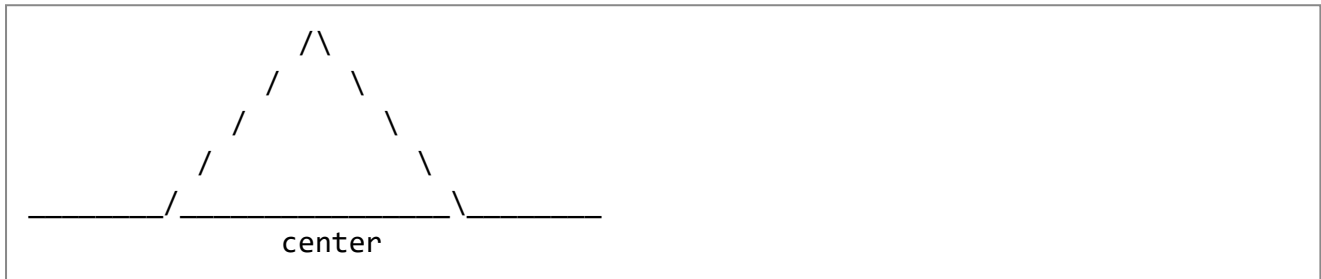


The center is around:

3

There are more observations around the center and fewer near the edges.

Now imagine many more observations producing a smoother shape:



That is the intuition behind the **bell curve**.

The horizontal direction represents possible values.

The height indicates where values are **more concentrated**.

16. Distributions are not always normal

It is important not to assume every dataset looks like a bell.

Consider:

1, 1, 1, 2, 3, 7, 15

Most observations are on the lower side, while 15 is far away.

Its shape might be uneven.

Another dataset could contain:

10, 10, 10, 10

There is essentially no spread at all.

So:

Distribution
≠
Automatically normal distribution

The normal distribution is simply one important pattern.

17. Easy edge cases

All values are equal

Consider:

5, 5, 5, 5, 5

Everything is at exactly the same location.

There is:

- one center,
- no variation,
- no meaningful bell-shaped spread.

Very small dataset

Consider:

10, 12

With only two observations, it is difficult to say much about the overall distribution.

A tiny sample may not reveal the real shape.

One value far away

Consider:

10, 11, 11, 12, 50

50 is far from the main group.

Most values are near:

10–12

while 50 lies far away.

The distribution will therefore not look nicely balanced around the central group.

18. Expected interpretation

For:

18, 19, 20, 20, 20, 21, 22

you should recognize that:

The center is around 20.

Values near the center include:

19, 20, 21

Values farther away include:

18, 22

A simple visual representation would be:



Therefore, the largest concentration appears around 20, while the number of observations decreases toward the edges.

The key idea for today is:

Probability distribution
↓
Shows how possible values are distributed

Normal distribution
↓
Many values near the center
Fewer values farther away
↓
Bell-shaped pattern

19. Hint only

For the exercise:

18, 19, 20, 20, 20, 21, 22

do not start with complicated mathematics.

First count how frequently each number appears:

18 → ?
19 → ?
20 → ?
21 → ?

Then imagine stacking one dot for every occurrence.

Ask yourself:

| Which number would create the tallest stack?

That tells you where the distribution is most concentrated.

Finally, compare values close to that center with values farther away. The important idea is **center versus spread**, not an advanced probability calculation.

By the way, ChatGPT Images 2.5 can turn a rough idea into a finished image, with richer textures and details you can refine. Want me to create an image of a beginner-friendly bell curve showing center, common values, and rare values?